Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	Blafar Ltd
Short name / acronym used in projects	BLAFAR
Participant Identification Code (PIC), if available	906676364
Website(s)	www.blafar.com
Country	Ireland
Organisation type	☐ Academic X Non-academic ☐ Hospital / clinic X SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Building 10, Cherrywood Business Park, Cherrywood, Loughlinstown Co. Dublin, D18 T3Y1 Ireland
Department(s), institute(s), research group(s) or unit(s) that would be involved	NA
Additional institutional resources relevant to collaboration	• Biomaterials synthesis and formulation laboratories • Injectable hydrogel design and characterisation platform • Functional biopolymer modification technologies • Quality management system supporting translational R&D (ISO 13485) • Access to advanced analytical facilities through University College Dublin, including polymer characterisation, rheology and materials analysis • Established collaborations with academic, clinical and industrial partners supporting translational biomaterials research

Main contact person for this profile	Dr Hongyun Tai, CEO and Managing Director
Main contact email	Hongyun.tai@blafar.com
Other contact person(s), if relevant	Prof Wenxin Wang, Founder & Chief Scientific Advisor

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☐ Retina
☐ Inherited retinal dystrophies (IRDs)	☐ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
x Advanced therapies / regenerative medicine	x Gene therapy / cell therapy / tissue engineering
☐ Inflammation / biomarkers / neuroprotection	x Drug delivery / formulations / nanomedicine
☐ Drug discovery / screening / repurposing	☐ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	Injectable hydrogels, Biomaterials engineering, Controlled drug delivery, Regenerative medicine, Tissue engineering, Biomaterial-enabled therapeutic delivery, Functional biopolymers, Ophthalmic biomaterials, Translational biomaterials, Advanced therapeutic medicinal products (ATMPs)
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus

Please describe the main focus of the team / unit that could participate. Keep it concise but specific (recommended: 150-250 words).

Blafar Ltd is an Irish research-intensive biotechnology SME developing advanced biomaterial platforms for regenerative medicine, advanced therapeutic medicinal products (ATMPs) and next-generation medical technologies. The participating team focuses on the design, synthesis and translation of functional biopolymers, injectable hydrogel systems and biomaterial-enabled therapeutic delivery technologies that address unmet clinical needs.

Our research integrates polymer chemistry, biomaterials engineering and translational product development to create biocompatible materials for controlled drug delivery, cell and gene therapy, tissue engineering and regenerative medicine. Particular expertise has been developed in chemically modified hyaluronic acid-based biomaterials and injectable hydrogel platforms with tunable mechanical properties, degradation profiles and therapeutic release kinetics.

The team actively collaborates with academic researchers, clinicians and industrial partners to accelerate the translation of biomaterial innovations from laboratory research towards clinical and commercial application. Current application areas include ophthalmology, musculoskeletal repair, wound healing and advanced drug delivery, while maintaining a platform technology approach that enables adaptation to a broad range of therapeutic indications.

Through participation in European collaborative research projects, Blafar aims to combine scientific excellence with industrial innovation, contributing expertise in biomaterial design, translational development and interdisciplinary

collaboration to support the development of next-generation healthcare technologies.

Why your team could be relevant to a translational eye-health and vision consortium

Please explain the specific added value your organisation / team could bring to a consortium centred on visual health, translational ophthalmology and related enabling technologies. ***Please think broadly: the contribution does not need to be directly ophthalmology-related. Interdisciplinarity is strongly encouraged, and expertise from apparently distant fields may open unexpected routes to innovation, new paradigms, and creative solutions to complex challenges that currently limit progress from research to patient benefit****.*

Blafar brings complementary expertise in advanced biomaterials, injectable hydrogel systems and translational product development that can enable innovative solutions for visual health beyond conventional ophthalmology research. Rather than focusing on a single disease area, our platform technologies are designed to support the controlled delivery of therapeutic molecules, cells and other advanced therapeutics, creating opportunities for interdisciplinary collaboration across regenerative medicine, drug delivery and tissue engineering.

Our team can contribute expertise in the design of functional biopolymers, biomaterial-assisted therapeutic delivery, formulation development and translation towards clinically relevant products. These capabilities are applicable to a wide range of ophthalmic challenges, including sustained intraocular drug delivery, regenerative approaches for ocular tissues, biomaterial-based cell and gene therapy delivery, and next-generation medical devices.

As an innovation-driven SME with experience in European collaborative projects, Blafar also contributes an industrial and translational perspective that complements academic and clinical research. We aim to help bridge the gap between laboratory discovery and practical healthcare solutions through technology development, regulatory awareness, quality-by-design principles and commercialisation planning.

By combining expertise in biomaterials engineering with close collaboration across academia, healthcare and industry, Blafar seeks to contribute enabling technologies that can accelerate innovation, foster interdisciplinary knowledge exchange and improve the translation of research into patient benefit.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

Please provide bullet points covering your strongest capabilities, methods, models, analytical approaches, translational assets or innovation strengths.

Technology Platforms

- Advanced biomaterials platform – Design and synthesis of functional natural and synthetic biomaterials for regenerative medicine and advanced therapeutic applications.
- Injectable hydrogel platform – Development of tunable hydrogel systems for sustained delivery of therapeutic molecules, cells and extracellular vesicles.
- Drug delivery and formulation technologies – Biomaterial-enabled delivery systems for small molecules, biologics and advanced therapeutics, with applications including ophthalmology.

Technical Expertise

- Functionalisation and engineering of hyaluronic acid and other biopolymers.
- Polymer synthesis, hydrogel fabrication and formulation optimisation.
- Physicochemical and rheological characterisation of biomaterials and hydrogel systems.
- Design of biomaterials with tunable mechanical properties, degradation profiles and controlled release characteristics.

Translation and Innovation

- Translation of biomaterial technologies from laboratory research towards clinically relevant products.
- Prototype development, design optimisation and manufacturability assessment.
- Experience in quality management, technology transfer, intellectual property and industrial collaboration.
- Development of biomaterial technologies for medical devices, regenerative medicine and advanced therapeutic medicinal products (ATMPs).

Knowledge Exchange and Collaboration

- Interdisciplinary collaboration between academia, clinicians and industry.
- Training in biomaterial design, hydrogel technologies and translational product development.
- Experience participating in European collaborative research and staff exchange projects.
- Support for innovation planning, commercialisation pathways and industry-oriented research translation.

Infrastructure, equipment, facilities or datasets relevant to consortium work

Examples: specialised laboratories, clean rooms, imaging platforms, -omics, biobanks, clinical cohorts, animal models, software tools, rehabilitation platforms, regulatory support or prototyping capacities.

- **Biomaterials Research Laboratory** for polymer synthesis, biomaterial functionalisation, hydrogel fabrication and formulation development.
- **Hydrogel and biomaterial characterisation capabilities**, including physicochemical analysis, rheological evaluation and formulation optimisation.
- **Prototype development capability** for injectable biomaterials, controlled therapeutic delivery systems and translational biomaterial technologies.
- **Quality management system (ISO 13485)** supporting research, technology development, documentation and innovation management for translational biomaterials.
- **Access to advanced analytical facilities through University College Dublin (UCD)**, including polymer characterisation, materials analysis and specialised instrumentation.
- **Access to academic and clinical collaboration networks** supporting in vitro evaluation, preclinical studies and application-specific validation of biomaterial technologies.
- **International industrial collaboration network** supporting technology translation, product development and future manufacturing scale-up.

Training capacity and knowledge-transfer potential

Please indicate what inbound staff could realistically learn in your organisation (scientific, technical, clinical, translational, regulatory, entrepreneurial or managerial skills).

Blafar provides a unique cross-sector training environment where researchers gain first-hand experience of the innovation pathway—from biomaterial design and laboratory development to quality-managed translational research and industrial collaboration.

Scientific and Technical Skills

- Design and functionalisation of natural and synthetic biomaterials.
- Injectable hydrogel formulation, optimisation and characterisation.
- Biomaterial engineering for controlled therapeutic delivery and regenerative medicine.
- Polymer synthesis, biomaterial modification and formulation development.

Translational Research and Innovation

- Translating biomaterial technologies from laboratory research towards clinically relevant products.
- Prototype development, design optimisation and technology validation.
- Integration of scientific research with industrial product development.

Quality Management and Regulatory Awareness

- Practical exposure to quality management systems supporting biomaterials research and innovation.
- Experience in design documentation, risk-based thinking and quality-by-design approaches for translational biomaterial development.
- Introduction to regulatory considerations and product development pathways for biomaterial-based medical technologies.

Industry–Academia Collaboration

- Working within multidisciplinary teams involving academic researchers, clinicians and industrial partners.
- Experience of collaborative R&D, European collaborative projects and cross-sector knowledge exchange.

What your team would hope to gain from participating
Examples: access to methods, models, technologies, patient-oriented validation, translational pathways, industrial insight, clinical exposure, data science, regulatory know-how, joint publications or longer-term strategic collaboration. ***Again, please think broadly and beyond ophthalmology.***
Blafar views participation in the consortium as an opportunity to strengthen long-term collaboration with leading academic, clinical and industrial partners in translational ophthalmology and regenerative medicine. We are particularly interested in expanding our understanding of ophthalmic disease mechanisms, preclinical models, clinical needs and translational pathways that will help guide the future development of our biomaterial platform technologies. The consortium would also provide valuable opportunities to identify new biomedical applications for Blafar's biomaterial platform technologies, validate biomaterial-enabled therapeutic delivery systems in clinically relevant settings, establish new interdisciplinary collaborations and explore emerging applications in ocular regeneration, advanced therapies and precision medicine. Through staff exchanges, we hope to foster long-term scientific partnerships, co-develop innovative research ideas, generate high-impact publications and prepare future collaborative European research proposals. We also value the opportunity for our researchers to gain exposure to complementary methodologies, clinical perspectives and translational research environments that will enhance both scientific excellence and industrial innovation.

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Dr Hongyun Tai	CEO				Hongyun.tai@blafar.com
Prof Wenxin Wang	CSO				Wenxin.wang@ucd.ie
Dr Rijiang Song	CTO				Rijiang.song@blafar.com
TBC	Product Development Scientist				
TBC	Product Development Scientist				

Main disciplines represented in your team	Biomaterials Science; Polymer Chemistry; Biomedical Engineering; Drug Delivery; Regenerative Medicine; Tissue Engineering; Materials Science; Quality Management; Technology Translation; Innovation Management.
Experience in EU / international / multi-partner collaborative projects	Blafar has extensive experience participating in European and international collaborative research projects involving academic institutions, SMEs, clinicians and industry partners. The company has contributed to multiple EU-funded collaborative programmes, including Marie Skłodowska-Curie Staff Exchanges, Horizon Europe and Enterprise Ireland-supported international projects, with expertise spanning biomaterials research, translational innovation, staff exchange, technology transfer and cross-sector collaboration.
Experience with clinical, translational or patient-facing research (if relevant)	x Yes ☐ No ☐ Not applicable | Short note: collaboration with clinical partners.

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements
Please prioritise achievements most relevant to your possible contribution. Please include DOI, URL or identifier.
1. Yu, H. et al.; Wang, W. 'MSC-derived exosomes injectable hyaluronic acid hydrogel for enhanced chronic wound healing'. JOURNAL OF CONTROLLED RELEASE, 2025, DOI: 10.1016/j.jconrel.2025.113985.

2. Chen, L. et al.; Wang, W. 'Biphasic release of betamethasone from an injectable HA hydrogel implant for alleviating lumbar disc herniation induced sciatica'. ACTA BIOMATERIALIA, 2024, 176, 173-189, DOI: 10.1016/j.actbio.2024.01.016.

3. Xu, Q. et al.; Wang, W. 'Versatile Hyperbranched Poly(β-hydrazide ester) Macromers as Injectable Antioxidative Hydrogels'. ACS Applied Materials & Interfaces, 2018, 10(46), 39494-39504. DOI: 10.1021/acsami.8b15006.

4. Wang, X. et al.; Wang, W. 'An injectable Chitosan-based self-healable hydrogel system as an antibacterial wound dressing'. Materials, 2021, 14(20), 5956. DOI: 10.3390/ma14205956.

5. He, Z. et al.; Wang, W. 'Reactive Oxidative Species (ROS): Utilizing Injectable Antioxidative Hydrogels and ROS-Producing Therapies to Manage the Double-Edged Sword'. Journal of Materials Chemistry B, 2021, 9, 6326-6346. DOI: 10.1039/D1TB00728A.

Up to 5 relevant previous projects, grants, networks or collaborations

Please provide detailed information (Title, Acronym, funding agency, GA-number, dates). Please indicate your role (coordinator / beneficiary / partner / service provider / collaborator). Indicate if international.

Note: GA numbers are not provided in the supplied CV/PDF for the entries below. Proposal number/call identifiers are included where available.

1. NET4MAT - Building knowledge and tools for the sustainable microbial fighting through sensing and responsive polysaccharide-based materials. Funding/programme: HORIZON-MSCA-2022-SE-01 / HORIZON-TMA-MSCA-SE. Proposal number: 101131205. Dates: 2024-2027. Role: Blafar Limited partner; Prof Wenxin Wang listed as a leading Blafar researcher. International: yes, 13-organisation consortium.

2. Drug Discovery and Delivery NEtwork for ONcology and Eye Therapeutics (3D NEONET). Funding/programme: EU Horizon 2020, H2020-MSCA-RISE-2016. Dates: 2017-2020. Role: Co-PI; €80,500 allocated to Prof Wang's sub-project from €945,000 total award. International: yes.

3. Novel Cell Migration Assay Based on Microtissue Technology and Tissue-Specific Matrices (MATRIXASSAY). Funding/programme: EU Horizon 2020, H2020-MSCA-RISE-2014. Dates: 2015-2019. Role: Co-PI; €54,000 allocated to Prof Wang's sub-project from €949,500 total award. International: yes.

4. Drug-Free Antibacterial Hybrid Biopolymers for Medical Applications (HyMedPoly). Funding/programme: EU Horizon 2020, H2020-MSCA-ITN-2014. Dates: 2014-2018. Role: Co-PI; €119,413 allocated to Prof Wang's sub-project from €3,918,055 total award. International: yes.

5. Novel combination of biopolymers and manufacturing technologies for production of a peripheral nerve implant containing an internal aligned channels array (NEURIMP). Funding/programme: EU FP7, FP7-NMP-2013-SME-7. Dates: 2013-2017. Role: Co-PI; €379,640 allocated to Prof Wang's sub-project from €3,304,705 total award. International: yes.

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)

Please indicate only what can be shared at this stage.

Know-how

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	x Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	x Only Researchers xTechnical Staff/Innovation x Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only x Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	Potentially, some of our staff need to apply for visa.

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice

Please summarise the 2-3 strengths that should be first visible in an internal searchable profile.

Blafar is an Irish translational biomaterials innovation SME specialising in advanced biomaterial platform technologies for regenerative medicine, advanced therapeutic medicinal products (ATMPs), therapeutic delivery and medical devices. Through close collaboration with academia, clinicians and industry, Blafar translates innovative biomaterials into clinically relevant healthcare solutions.

Capabilities or expertise you would especially like to access through the consortium
Please indicate the types of teams, methods, infrastructures or translational pathways you would like to connect with in the consortium. Blafar would particularly welcome collaboration with teams offering complementary expertise in ophthalmic biology, disease modelling, preclinical evaluation and clinical translation. We are interested in connecting with researchers and clinicians who can help validate biomaterial platform technologies in clinically relevant models and identify unmet clinical needs that can drive future technology development. We are especially interested in collaborating with partners working in: • Ophthalmic disease models and preclinical validation. • Ocular drug delivery, regenerative medicine and advanced therapeutic development. • Cell and gene therapy, extracellular vesicle (EV) therapeutics and tissue engineering. • Clinical translation, regulatory science and patient-oriented evaluation. • International industrial–academic collaborations that support technology translation and future collaborative research.

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	x Yes ☐ No
x We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
NA

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

x Understood

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Date completed: 05/07/2026

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